



Bayan

On-device Arabic text simplification for readers with dyslexia

Team Cogni

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Arabic hides its own pronunciation

11%

of Arab primary-school children
have developmental dyslexia

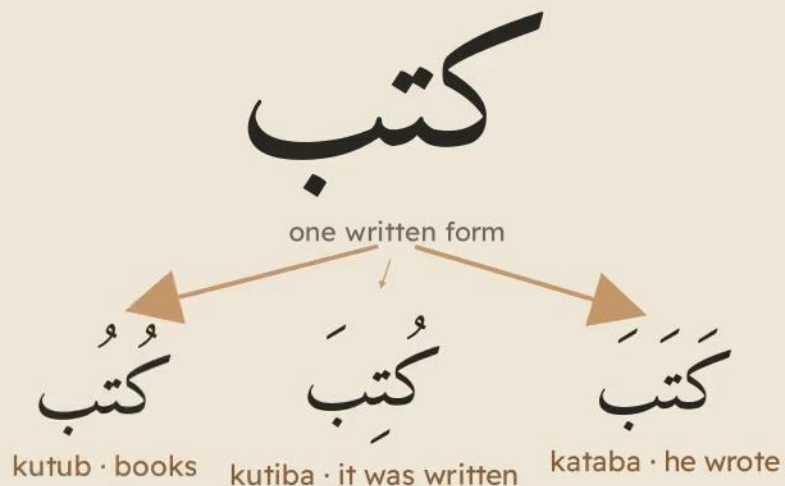
pooled over 18 studies, N = 30,243

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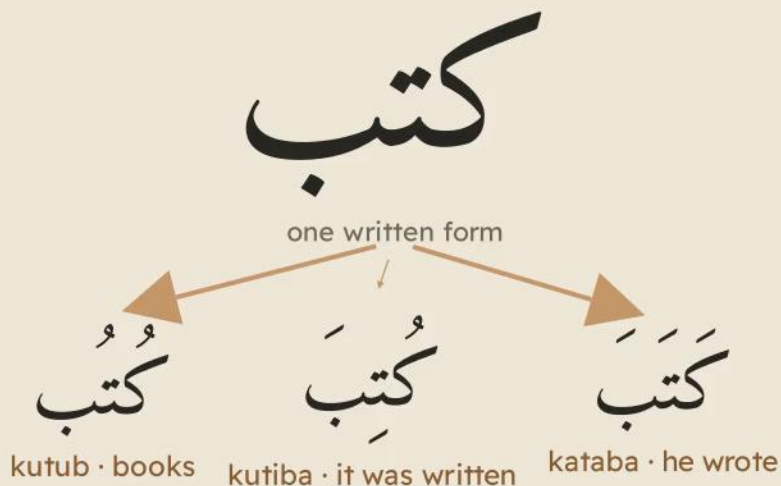


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Fused prefixes and suffixes → long words that resist chunking

Clauses chained with و and ف → sentences longer than working memory

Nobody had built this: no tool, no data

	SAMER 15 novels, 3 levels	BAREC 69,441 sentences	Baseet 3 LLM levels	DAASI gov. & insurance text
Rewrites by	human	none (levels only)	LLM	human-verified
Adds a sentence	0% of pairs	n/a	2-4%	8%

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No deployed Arabic dyslexia tool simplifies text, and no open corpus teaches splitting.

- So we built our own data, our own models and our own benchmark.

Simplify where the reader already is

Where does a reader meet hard Arabic? Inside other apps.

A

Chrome extension

desktop-first; reading happens on the phone

B

Android plugin

in the selection menu of every app 'تبسيط'

C

Standalone app

copy, paste, leave the app: the friction we remove

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on-device

offline

≤ 250 MB

the reader's text never leaves the phone



Audit first, then build the training set

9.3%

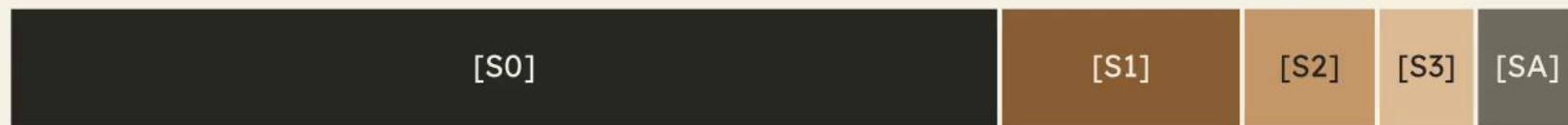
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22,733 training pairs, every one tagged with its source style



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SAMER · 6,033 unchanged kept

■ 3,508

Baseet · light

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Baseet · medium

■ 1,426

Baseet · strong

■ 1,502

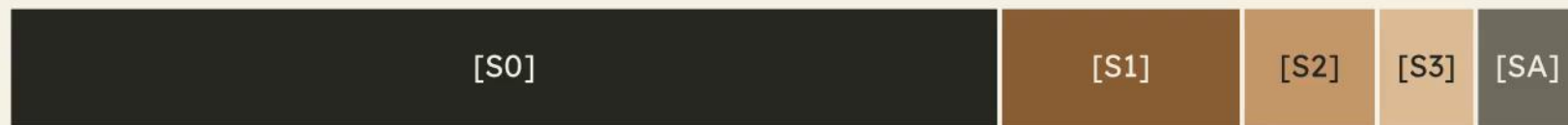
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- Baseet pairs pass a meaning filter: $\text{LaBSE} \geq 0.646$ · $\text{length} \geq 67\%$ of source · every number kept
each threshold = 5th percentile of human-verified rewrites (DAASI)

Two candidates, one recipe

one script (train.py) · identical data · architecture is the only difference

10 epochs

fp16

seed 42

max length 256

best SARI on changed rows

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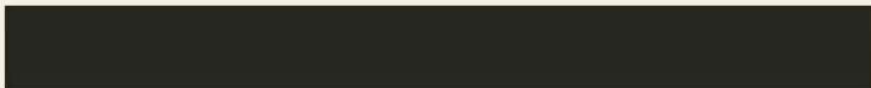
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AraT5v2



368M params · won Baseet's head-to-head · models 1 & 2

AraBART



139M params · 2.6× smaller · the on-device contender

HPLT T5



≈ 294M params · held in reserve

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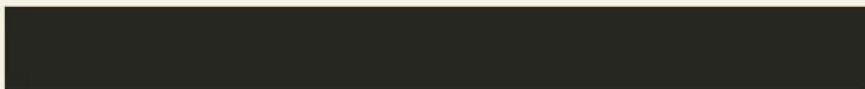
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- 46% of AraT5v2 is two embedding tables, so shrinking its vocabulary is the route to phone size.

Model 1: the score that looked like a failure

SARI on all SAMER test rows
3,277 sentences, half need no change



Model 1: the score that looked like a failure

SARI on rows a human changed

1,678 sentences: the bar a model must clear



But it made mostly one-word edits, and 42% of its training pairs were unchanged sentences.

Model 2: one fix for each lesson

It rewards doing nothing



Keep all 6,033 unchanged pairs;
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[S0]

minimal

[S1]

light

[S2]

medium · default

[S3]

strong

[SA]

everyday



Simplify where you read.



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Next: does it work?

BayanBench: a benchmark for the reader's task

On a real phone

Readers with dyslexia

Together for Tomorrow!
Enabling People

Education for Future Generations